SUPERSTORE SALES

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ANALYTICAL OBJECTIVES

- Product insight: Understanding products performance.
- Sales performance: Understanding sales performance.
- Operations: Understanding orders and delivery efficiency.
- Customer insight: Understanding customer segment and behaviour across sales region.

REASEACH QUESTIONS

1. Which product category contributed more to total sales ?
2. Did product experience any difficulties in sales or experience loss?
3. How does sales trended over time?
4. What is the average delivery time to deliver a product and is delivery period affected by shipment mood?
5. Which shipment mood is the most used?
6. How did customer segment performed in each region
7. Which quarter does customer segment generated more sales?

Understanding dataset, limitation cleaning methodology and preparation

This is a super store sales data from Kaggle and it consist of 9994 rows and 21 columns, like order_id order date, ship date, product_id, product cat, sub-cat, region, state, city customer name, segment, sales, discount, quantity, profit..

Going through the data I discovered the data was a little bit messy, columns like sales, profit, discount, quantity had wrong data type formatting. Fixing this I loaded the data to **power query** and applied consistent data type formatting, I formatted sales, profit, columns from numeric to currency.

Another observation was that the profit column has negatives number which I assumed were losses but to be quite sure I created cost column and cost per item column, I also added a profit status column, expected profit column, and profit difference column to track and identify product with loss, to track product that “performed lower” or “ higher than expected” I added a profit accuracy column.

Understanding dataset, limitation, cleaning methodology and preparation

Finally I created a shipment duration column from the order date and ship date column to track how long delivery process take.

Also during the cleaning process I realize the discount field stored rate value (e.g, 0.2 instead of 20%). So I reformatted the column to percentage for clarity and consistency.

Data cleaning/preparation tool: Excel, power query..

ANALYSIS METHOD

- Descriptive; This was used to describe my numerical data such as, sales, profits, discount, cost, quantity
- Comparative: This method was used to compare groups using pivot table, and charts.
- Trends: This was used for sales analysis and customer insight analysis, (e.g sales trend over time & customer segment sales trends)

Analytical Dashboard

TOTAL SALES

$11,488,062

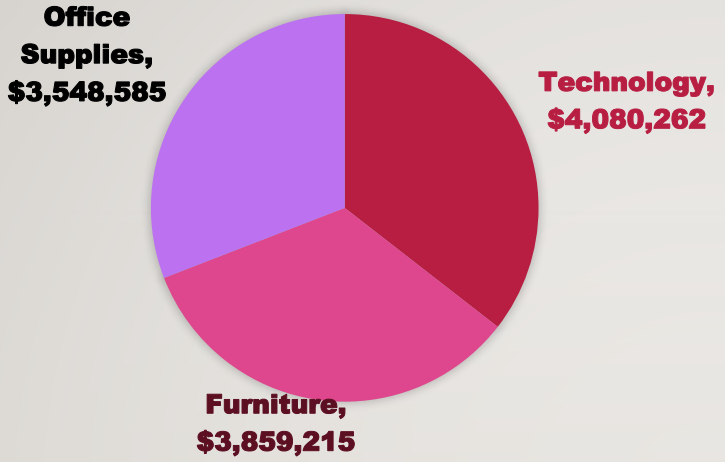
TOTAL PROFIT

$286,397

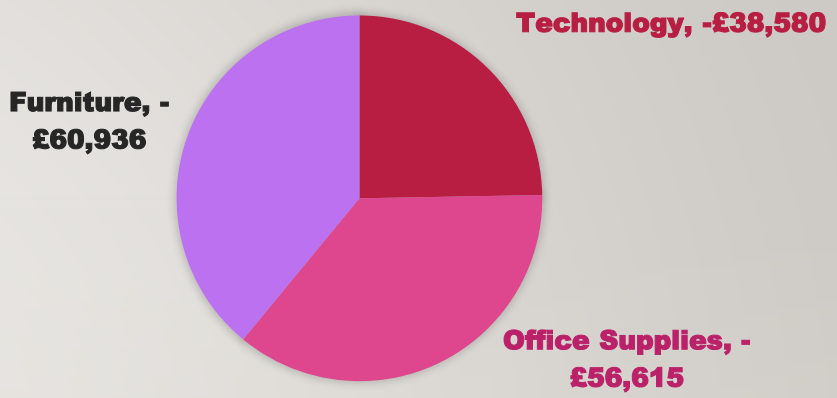
AVG DELIVERY DAY

4

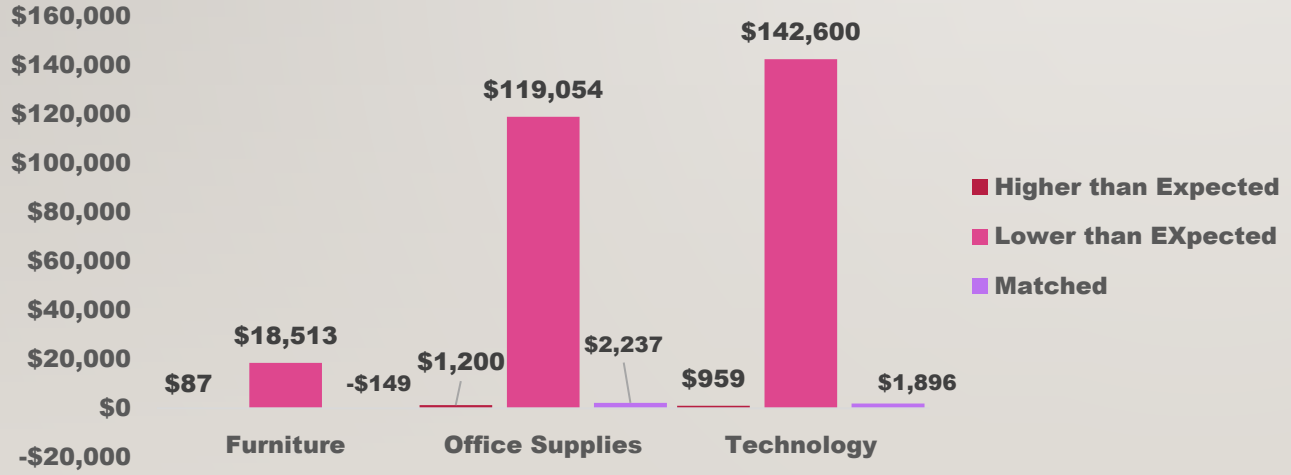
PRODUCT CAT WITH MOST SALES



PRODUCT CAT WITH MOST LOSS

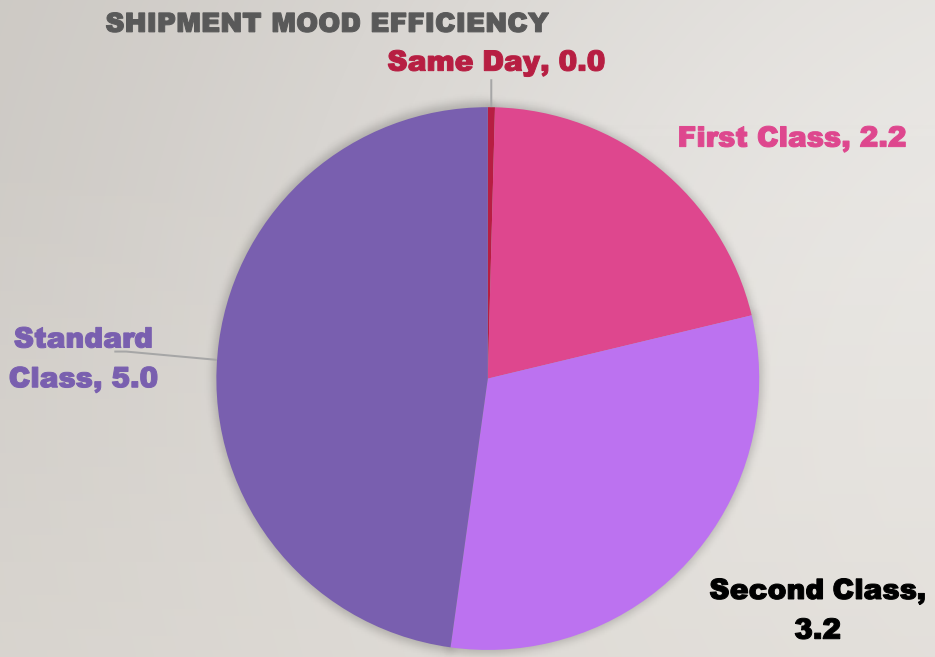


Performance Accurancy

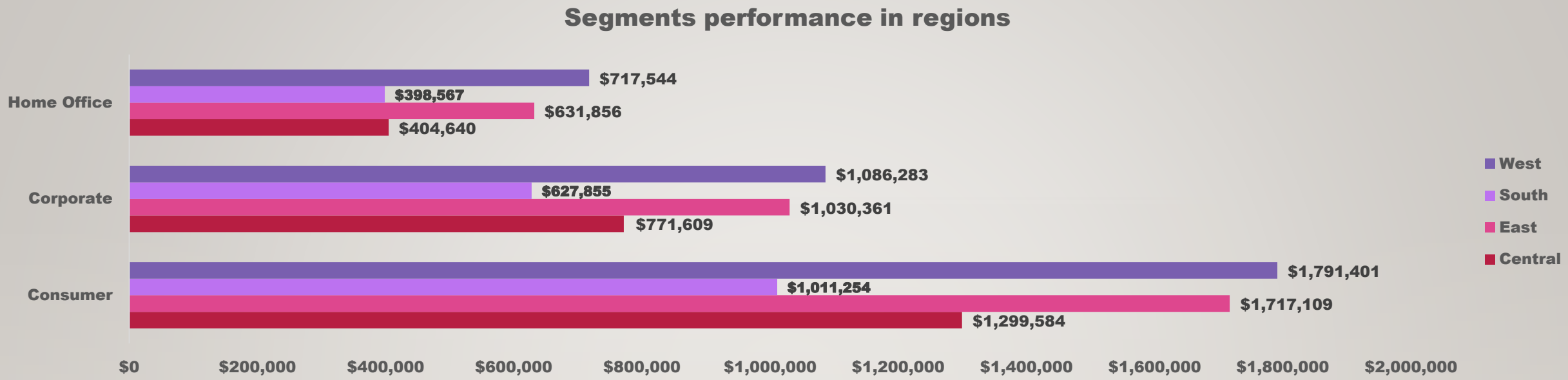


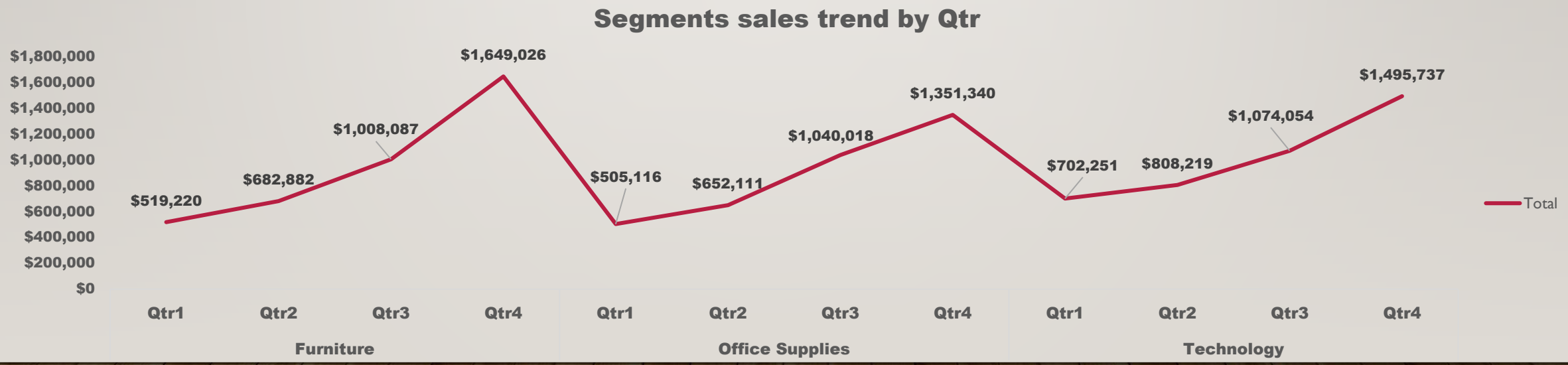
Sales trend over time

Analytical Dashboard



Analytical Dashboard





Insights & Suggestion

From the Analysis Technology category contributed more to total sales, with a total of $4,080 even though it performed lower than expected. However all categories underperformed with actual profits significantly lower than projections, While some loss-making categories performed better than expected, reflecting gradual recovery, the general pattern of underperformance across categories highlights a need to reassess pricing strategies and cost structures. Focusing on optimizing discount policies and improving cost efficiency could help narrow the gap between expected and actual profits in future sales period. Sales between 2011/12 wasn't so great probably due to the fact that the superstore started sales this period, but there's was a significant increase in sales from 2013/2014 which may be as a result of longevity and customer satisfaction with products which further indicates growth. Same day delivery mood appeared to be the fastest delivery process, although customers prefer to use the standard delivery mood which may be as a result of price sensitivity. Customer segment performed most in every region ahead of corporate and home office segments, customer segments experienced a very high sales in the western and central regions. Cooperate segment didn't do so badly as it also had significant sales in each region but home office experience fewer sales in each region which maybe as a result of home office being price sensitive and buy fewer high value items, also this may be affected by marketing and targeting, marketing efforts may be focused on customer and cooperate segment, so better marketing efforts should be equally channeled to home office segments to reach it customers in these regions. Sales across all segments show a clear upward trend through the year, with a consistent dip in Q1 and strong recovery peaking in Q4. Furniture and Office Supplies recorded steady growth, rising from around $500K in Q1 to over $1.3M in Q4, reflecting strong year-end demand. Technology led overall performance, climbing from $702K in Q1 to $1.49M in Q4 — the highest Q4 value among all segments. This pattern suggests seasonal purchasing behavior, where spending intensifies toward the end of the year, possibly due to budget utilization and business expansions. Q4 remains the key revenue-driving period, while Q1 slowdowns may indicate post-holiday demand drops or procurement pauses.